THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4	Four C–H bonds
2	**(a)** –OH	Alcohol
3	**(d)** ethanoic acid	Two-carbon acid
4	**(b)** Incorrect — both are allotropes of carbon	Allotropes
5	**(a)** Student A is correct	Alkynes: C_nH_{2n-2}
6	**(a)** C_3H_8	Alkane
7	**(a)** hydrogenation	Addition reaction
8	**(c)** esterification	Ester formed
9	**(c)** ethanoic acid	Oxidation
10	**(b)** catenation	Self-linking

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gaining or losing 4 electrons needs too much energy, so carbon shares electrons.	2
12	Each N shares three electron pairs — a triple bond N≡N with one lone pair on each N.	2
13	Clusters of soap molecules in water with hydrophobic tails inside and ionic heads outside, trapping oily dirt.	2
14	False — soaps react with Ca^{2+} and Mg^{2+} ions to form insoluble scum; detergents work in hard water.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A series of compounds with the same functional group and similar properties; successive members differ by CH_2 (14 u); same general formula; gradual change in physical properties. e.g. CH_4, C_2H_6, C_3H_8	1+2
16	(i) $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ + heat (ii) $CH_4 + Cl_2 \xrightarrow{\text{sunlight}} CH_3Cl + HCl$ (iii) $CH_2{=}CH_2 + H_2 \xrightarrow{\text{Ni}} CH_3{-}CH_3$	1+1+1
17	Compounds with the same molecular formula but different structures; n-butane $CH_3{-}CH_2{-}CH_2{-}CH_3$ and isobutane $CH_3{-}CH(CH_3){-}CH_3$	1+1+1
	OR: (i) Add $NaHCO_3$: ethanoic acid gives brisk effervescence of CO_2, ethanol does not. (ii) Blue litmus turns red with ethanoic acid only.	1½+1½
18	Soap molecule: ionic head (hydrophilic) and hydrocarbon tail (hydrophobic). Tails attach to oil, heads face water forming micelles; rinsing removes the dirt.	3
	OR: (i) complete combustion (ii) incomplete combustion due to higher carbon content (iii) to provide enough oxygen for complete combustion	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) catenation — forms long chains, branches, rings; tetravalency — bonds with 4 atoms incl. O, N, S, Cl (ii) $CH_3{-}C(=O){-}OH$ (iii) aldehyde (–CHO); ketone (C=O)	2+1+2
	OR: (i) $CH_3COOH + C_2H_5OH \xrightarrow{\text{acid}} CH_3COOC_2H_5 + H_2O$ (ii) alkaline hydrolysis of ester to give alcohol and sodium salt of acid: $CH_3COOC_2H_5 + NaOH \longrightarrow C_2H_5OH + CH_3COONa$ (iii) perfumes/flavouring	2+2+1
20	(i) X: ethanol, Y: ethene (ii) $CH_3CH_2OH \xrightarrow[443\,K]{\text{conc}\cdot H_2SO_4} CH_2{=}CH_2 + H_2O$ (iii) dehydrating agent (iv) $2\,Na + 2\,CH_3CH_2OH \longrightarrow 2\,CH_3CH_2ONa + H_2$	1+2+1+1
	OR: (i) soaps: Na/K salts of long-chain carboxylic acids, form scum in hard water, biodegradable; detergents: ammonium/sulphonate salts, work in hard water, often non-biodegradable (ii) non-biodegradable ones persist (iii) water containing Ca^{2+}/Mg^{2+} salts	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4	Atomic no. 6 → 2, 4
2	**(d)** C_5H_{10}	Starts from C_2H_4
3	**(d)** acetic acid	Vinegar is 5–8% solution
4	**(b)** Incorrect — only unsaturated compounds do	Ethane is saturated
5	**(b)** Student A is correct	m.p. 290 K
6	**(d)** 7	6 C–H + 1 C–C
7	**(b)** fullerene	C_{60} buckminsterfullerene
8	**(b)** –al	Propanal
9	**(d)** hydrogen	Sodium ethoxide formed
10	**(a)** ethyne	Acetylene

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Carbon has four valence electrons and forms four covalent bonds.	2
12	C_2H_5OH (ethanol) and C_3H_7OH (propanol)	1+1
13	Each carbon is bonded to three others; the fourth electron is free (delocalised).	2
14	False — methanol is oxidised to methanal in the liver, which can cause blindness and death.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) C–C single bond, 3 H each (ii) C=C double bond, 2 H each (iii) O=C=O with two double bonds and lone pairs on O	1+1+1
16	Unsaturated compounds add atoms across double/triple bond, e.g. $CH_2{=}CH_2 + H_2 \xrightarrow{Ni} C_2H_6$; hydrogenation of vegetable oils	1+1+1
17	An atom/group of atoms giving characteristic properties; (i) aldehyde, ethanal (ii) ketone, propanone	1+1+1
	OR: Ethanol mixed with methanol/poison and dye to make it unfit for drinking; to prevent misuse of industrial alcohol; slows metabolism, causes lack of coordination	1+1+1
18	Soap anions react with Ca^{2+}/Mg^{2+} forming insoluble scum (wasting soap); detergent's charged ends do not form insoluble precipitates.	3
	OR: (i) $CH_3CH_2CH_2CH_2CH_2Br$ (ii) $CH_3COCH_2CH_3$ (iii) $CH_3(CH_2)_4CHO$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $CH_3CH_2OH \xrightarrow{alk\cdot KMnO_4+heat} CH_3COOH$ (ii) substances that add oxygen: alkaline $KMnO_4$, acidified $K_2Cr_2O_7$ (iii) combustion is burning in oxygen with heat and light; oxidation here converts alcohol to acid without burning	2+2+1
	OR: (i) $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$; $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$; $CH_3COOH + C_2H_5OH \xrightarrow{acid} CH_3COOC_2H_5 + H_2O$ (ii) it ionises partially	3+2
20	(i) P alkane, Q alkene, R alkyne (ii) Q and R (iii) $CH_3{-}CH{=}CH_2$ (iv) P	2+1+1+1
	OR: (i) formed by sharing electrons (ii) no ions; weak intermolecular forces (iii) O shares one pair with each H with two lone pairs; N shares with 3 H with one lone pair	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.